

# THE FIVE CLOCKS

京张五速

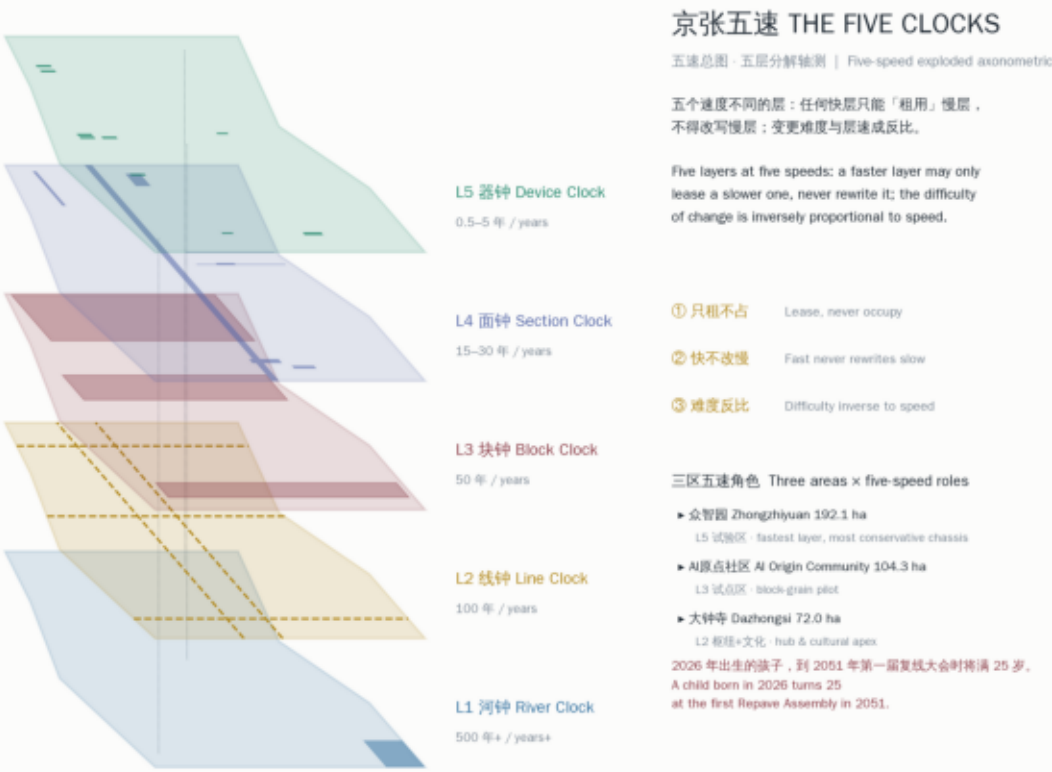
A Chassis for the Next Hundred Years

# Executive Summary

## 执行摘要

This belt consists of five layers moving at different speeds. Each renews on its own cycle; a faster layer may only lease a slower one, never rewrite it; the difficulty of change is inversely proportional to speed.

- The mismatch: AI turns over in 0.5–2 years; the chassis must last 100
- Method: solid slow layers + standard interfaces for fast layers
- Site law: the alignment outlives every train (1905–2026)
- Deliverable: 5 layers + 3 laws + inverse procedures + 25-y Assembly
- A child born in 2026 turns 25 at the 2051 Assembly



概念示意 · 临时边界 · 非官方红线 | Conceptual · provisional boundary · NOT an official redline | assets/figures/site-overview.png | 《京张五速 THE FIVE CLOCKS》 submissions/tyro/five-clocks · CC-BY-SA-4.0

# Method: Century Design ≠ Century Prediction

方法论：百年设计 ≠ 预测百年

|                         |                                               |
|-------------------------|-----------------------------------------------|
| 康泽恩 Conzen 1960         | Streets outlive plots outlive buildings       |
| 布兰德 Brand 1994          | Six shearing layers at different paces        |
| 罗西 Rossi 1966           | Permanences that withstand time               |
| 哈布瑞肯 Habraken 1961      | Supports separated from infill (SI)           |
| 北京总规 Beijing masterplan | Strategic reserve: not deciding is a decision |



Shared conclusion: build the un-guessable layers solidly; give the guessable layers standard interfaces. This proposal refuses to predict the technology of 2126 — the refusal is the method.

- Three laws: lease not occupy / fast never rewrites slow / inverse difficulty
- "The robots need it" is never a reason to modify the street network
- L5 default exit: inaction removes the device

# The Five-Layer Matrix

五层矩阵

| Layer/Clock   | Period*   | Instrument                                   | Procedure                        |
|---------------|-----------|----------------------------------------------|----------------------------------|
| L1 河钟 River   | 500 年+ y  | Blue-green protection line<br>蓝绿保护线          | Not amendable<br>不可议             |
| L2 线钟 Line    | 100 年 y   | Alignment-protection plan<br>线位保护图则          | 25-y Repave Assembly<br>25 年复线大会 |
| L3 块钟 Block   | 50 年 y    | Grain guideline + easement<br>颗粒度导则+地役权      | Public hearing<br>听证             |
| L4 面钟 Section | 15–30 年 y | Elastic sections + device band<br>弹性断面图集+设备带 | Planning permit<br>规划许可          |
| L5 器钟 Device  | 0.5–5 年 y | Lease + ledger + deposit pool<br>租约制+账牌+押金池  | Filing · default exit<br>备案·默认退出 |

\* Periods are design assumptions (A-CLOCK-001). The naming echoes Dazhongsi: the Yongle Bell tolls hours; these clocks toll eras.

Diagnosis: Grain and Sections

诊断：颗粒与断面

The double cut: the R-02 census has been executed (2026-08-21 OSM snapshot, ODbL). The belt is cut twice — lengthwise by at-grade rail, crosswise by compounds. Topology basis, not legal enclosure; provisional bbox.

- ▶ 344 blocks, bimodal: median 0.662 ha; 66 superblocs (19.2%) hold 84.8% of area
- ▶ At-grade Jingbao line: 8 crossings over 6,504.7 m (1/813 m); max gap 1,110.2 m
- ▶ 3 reserved corridors target the worst 3.24 km band (2 crossings only)
- ▶ The L4 lesion: aged sections without device band or conduit headroom
- ▶ Renewal: retain first, demolish the exception (Beijing regulation)

指标—证据链 Metrics—Evidence Chain



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证据链 Evidence chain



R-02 街坊颗粒普查协议 Block-grain census protocol



三项视觉核验指标 Three visual-gate metrics



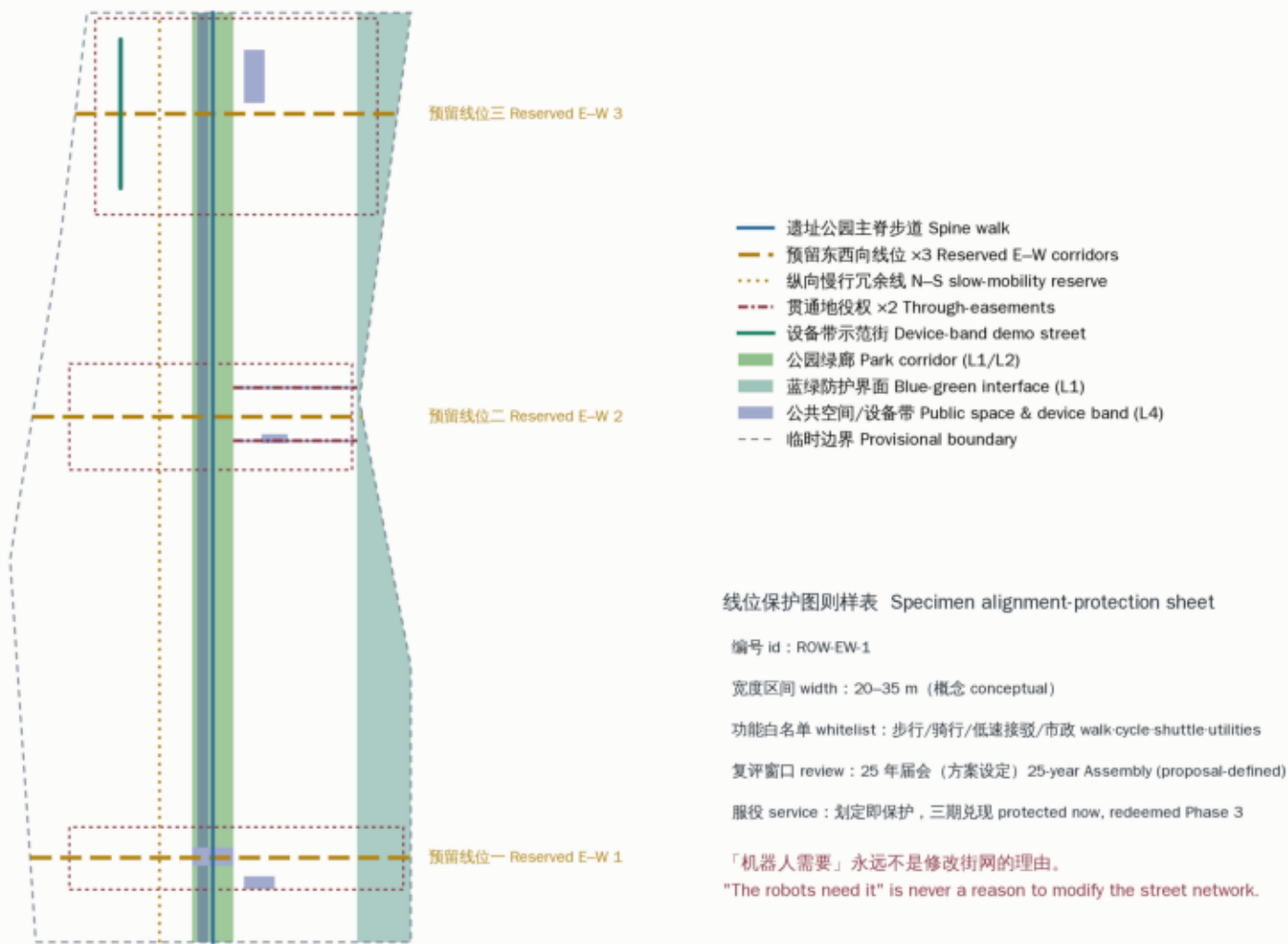
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Recomputed from this package's geometry in EPSG:4548; provisional-boundary basis, low confidence.

第一指标 slow\_layer\_change\_count = 0 (恒)  
The first metric: slow\_layer\_change\_count = 0, always.

# The Alignment Atlas

## 线位图则

### 交通慢行与蓝绿复合系统 Mobility & Blue-Green Composite



线位保护图则样表 Specimen alignment-protection sheet

编号 id : ROW-EW-1

宽度区间 width : 20–35 m (概念 conceptual)

功能白名单 whitelist : 步行/骑行/低速接驳/市政 walk-cycle-shuttle-utilities

复评窗口 review : 25 年届会 (方案设定) 25-year Assembly (proposal-defined)

服役 service : 划定即保护, 三期兑现 protected now, redeemed Phase 3

「机器人需要」永远不是修改街网的理由。  
"The robots need it" is never a reason to modify the street network.

L1 蓝绿 × L2 线位 × 预留线位叠合 (临时边界) Blue-green × alignments × reserved corridors (provisional)

概念示意·临时边界·非官方红线 | Conceptual · provisional boundary · NOT an official redline | assets/figures/mobility-bluegreen.png | 《京张五速 THE FIVE CLOCKS》 submissions/Xyro/five-clocks · CC-BY-SA-4.0

► 3 reserved E–W corridors: Dazhongsi / Origin / Zhongzhiyuan

► 1 N–S slow-mobility redundancy line, parallel to the spine

► Per sheet: id / width range / whitelist / service dates / review window

► Review window: the 25-year Assembly only (proposal-defined mechanism)

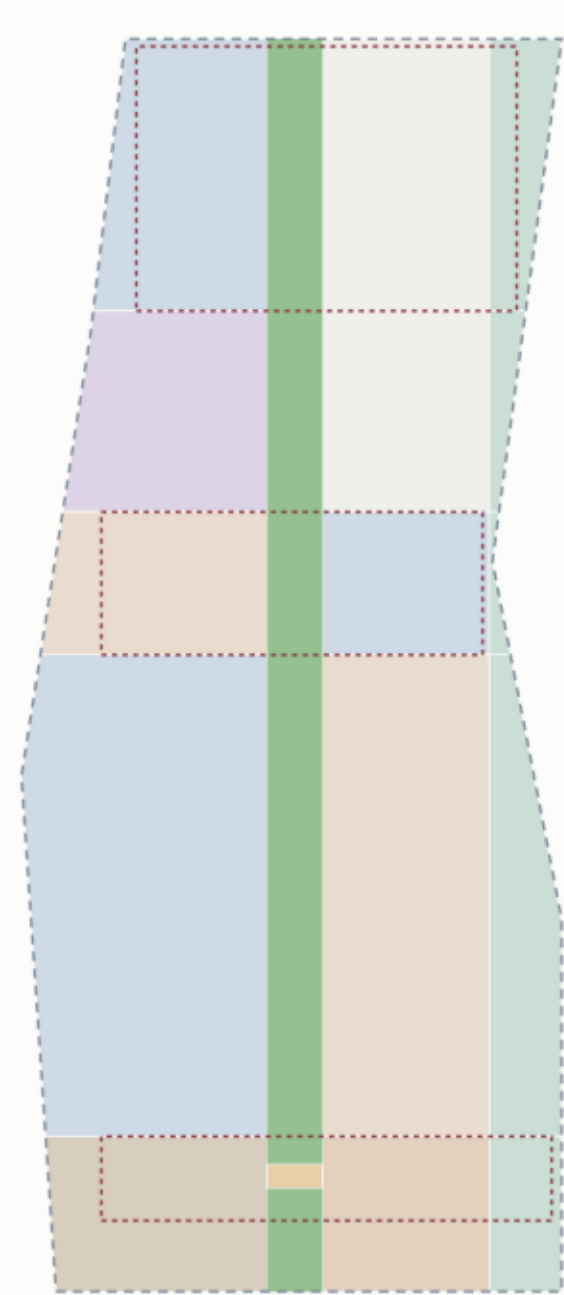
► All conceptual corridors, not engineering alignments (A-ROW-001)



# Layered Control Guidance

分层管控引导

## 分层管控引导图 Layered Control Guidance



用地图例 Land-use legend

- 0802 科研 Research
- 0701 居住 Residential
- 0804 教育 Education
- 0902 商务 Business
- 0901 商业 Commercial
- 16 留白 Reserve
- 1401 公园绿地 Park green
- 1402 防护绿地 Protective green
- 1403 广场 Plaza

### 分层管控引导一览 Layered control guidance

每条指标 = 管控内容 + 锁定层位 + 复评周期  
Each indicator carries a lock layer and a review period

|                                 |                                                |
|---------------------------------|------------------------------------------------|
| 蓝绿保护线 Blue-green line           | 锁定 lock: L1 复评 review: 不可议 Not amendable       |
| 线位与通行权 Alignments & ROW         | 锁定 lock: L2 复评 review: 25 年届会 25-y Assembly    |
| 贴线率/街墙 Build-to & street wall   | 锁定 lock: L3 复评 review: 50 年 50 y               |
| 高度分区骨架 Height zoning skeleton   | 锁定 lock: L3 复评 review: 50 年 50 y               |
| 街坊颗粒上限 Block-grain ceiling      | 锁定 lock: L3 复评 review: 50 年(听证) 50 y (hearing) |
| 屋顶/第五立面 Roof & fifth facade     | 锁定 lock: L4 复评 review: 30 年 30 y               |
| 弹性断面/设备带 Sections & device band | 锁定 lock: L4 复评 review: 15–30 年 y               |
| 广告与临时装置 Ads & installations     | 锁定 lock: L5 复评 review: 年度 Annual               |

容积率/高度/密度/绿地率/退线 : 待正式控规条件公布  
FAR/height/density/green ratio/setback: pending official controls

Every control indicator = content + lock layer + review period.

FAR/height etc. await official conditions; no approved values.

用地镶嵌概念模型（临时边界） Conceptual land-use mosaic (provisional)

# Zhongzhiyuan: Fastest Layer, Most Conservative Chassis

众智园：最快的层，最保守的底盘

- The L5 test zone · 192.1 ha (provisional)
- Speed segregation: an L2-grade green corridor separates test from public
- However wild the tests, the chassis does not move
- Test transcripts price public-street lease deposits (S01 channel)
- Chassis redundancy: conduit headroom + elastic sections + device band

## 三区五速角色 Three Key Areas × Five-Speed Roles



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# Zhongzhiyuan: The Device-Band Demonstration Street

众智园：设备带示范街

- The belt's first full device-band street, ~1.55 km (R-04)
- A 0.8–1.5 m reversible curbside strip: the only legal L5 landing
- Ledger plates: layer / term / deposit / restoration / parity path
- Unleased for two cycles → restored (the street obeys default exit too)
- Annual student co-creation leases (persona P5's interface)

Living-street elastic section (conceptual)



# Origin Community: Winding the Block Clock

原点社区：拨动块钟

► The L3 pilot · 104.3 ha (provisional)

► Grain subdivision of an unnamed campus cluster (illustrative)

► Never compels stock; rights-holders decline → next candidate (R-06)

► Two through-easements (Civil Code arts. 372–385)

► Youth housing on new blocks: the L3 dividend to the youngest

Before: one 30-ha low-permeability compound

After: 4–6 small blocks + two 24 h passages  
(illustrative; no real compound)

# Origin Community: The Lease Service Station

原点社区：租约服务站

- ▶ Ledger plates, deposits, filings in one stop: the citizen-visible institution
- ▶ Deposit custody: licensed institution + regulatory compliance (pending)
- ▶ Five lease elements: layer/term/deposit/restoration/parity path
- ▶ Default exit: no renewal = automatic removal
- ▶ New professions: lease examiners, Assembly secretariat (slow-layer services)

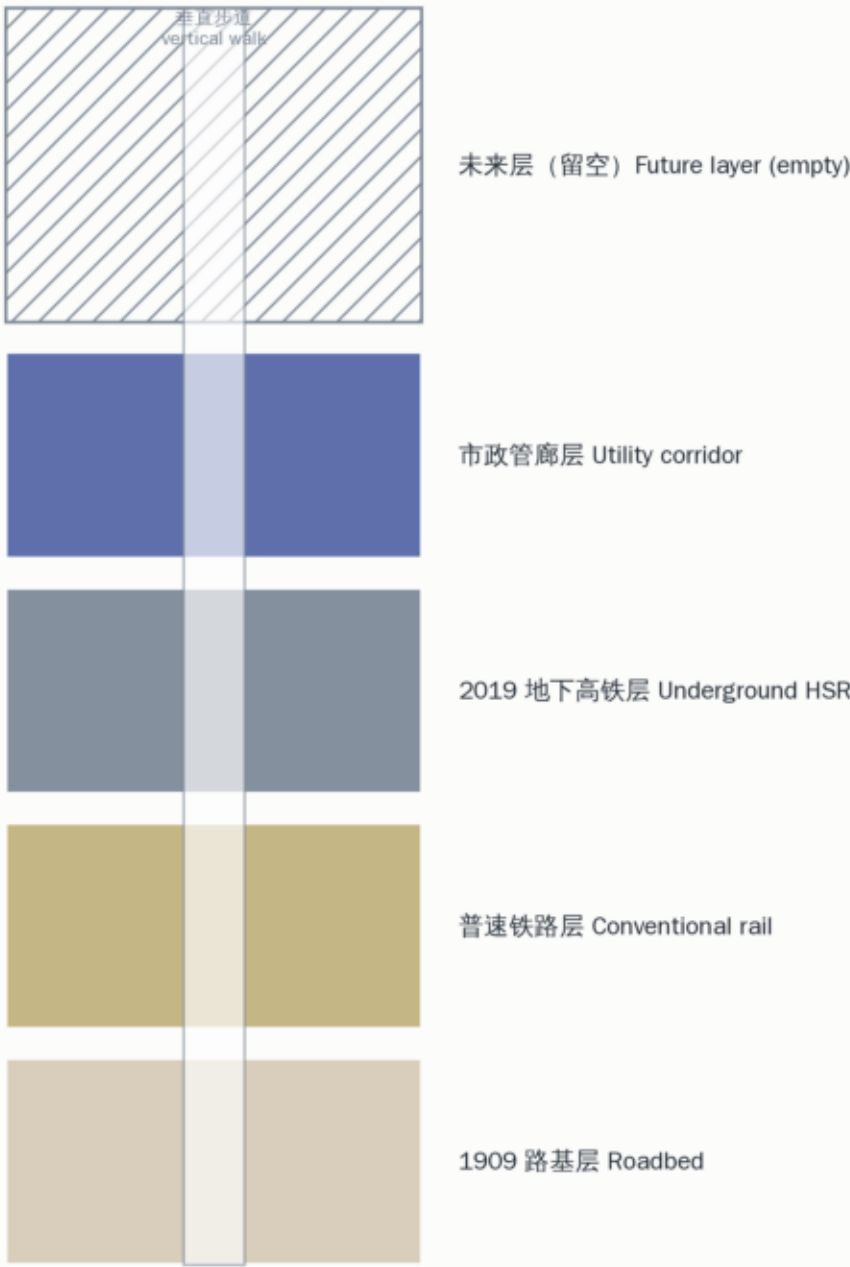
# Dazhongsi: The Section Tower

大钟寺：断面塔

- ▶ An enterable vertical museum: one alignment, six technology strata
- ▶ The top "future layer" left deliberately empty
- ▶ Curatorial concept: no engineering feasibility conclusions
- ▶ Supports kept + curatorial infill (the King's Cross reading)

## 断面塔（概念剖切） Section Tower (conceptual cutaway)

策展性构筑物：不含桥隧/地下工程可行性结论  
Curatorial structure: no engineering feasibility conclusions



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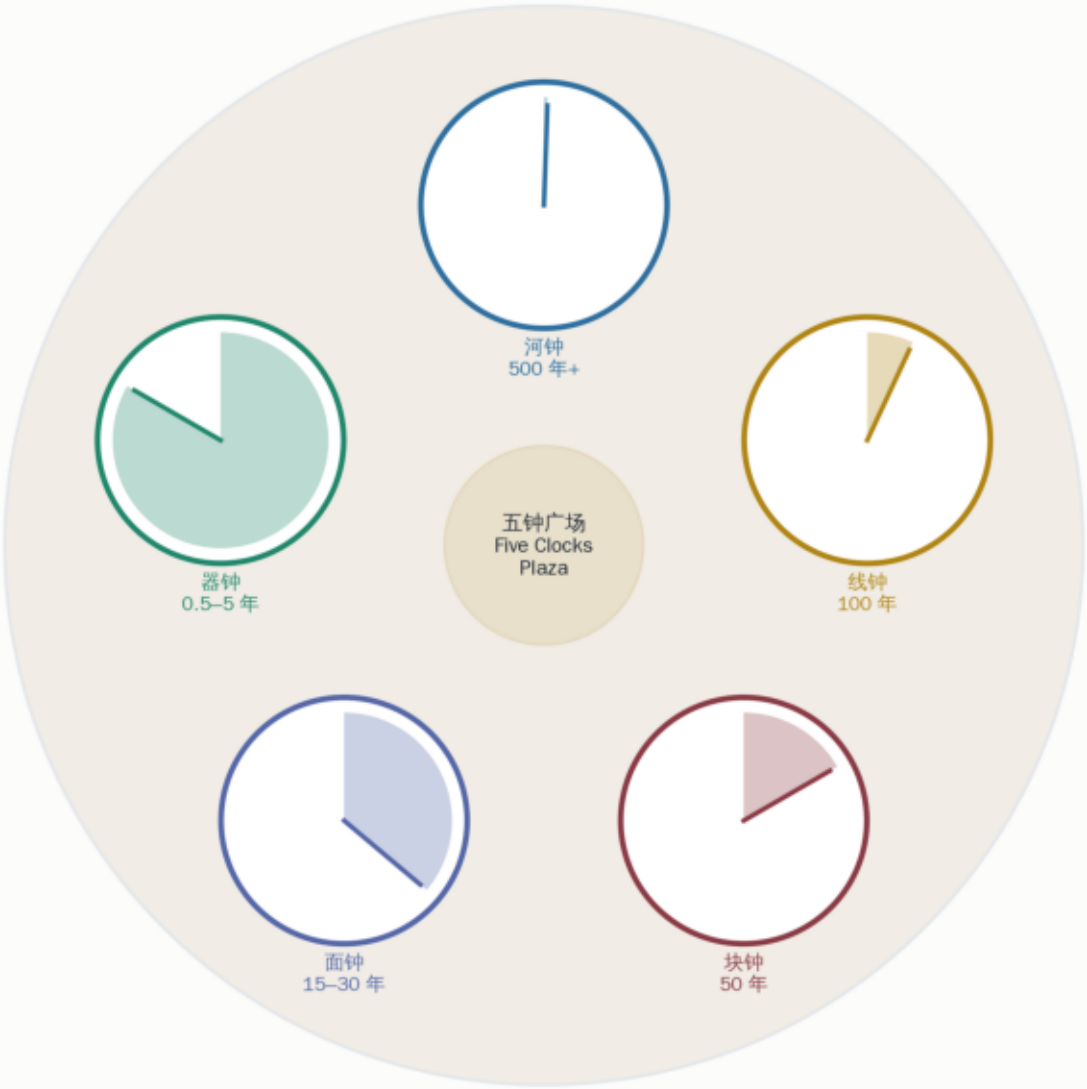
# Dazhongsi: Five Clocks Plaza

大钟寺：五钟广场

- Five city clocks, each at its layer's speed
- The Device dial spins; the River dial barely moves
- Public art + science communication, never entertainment
- Timeline paving terminus: a year per 50 m; future years empty
- Covenant stele: honor rings, one per 25 years

五钟广场平面（概念） Five Clocks Plaza plan (conceptual)

五口城市之钟各按层速运转：器钟快转，河钟近乎不动。公共艺术+科普定位。  
Five city clocks, each at its layer's speed: the Device dial spins; the River dial barely moves. Public art + science communication.



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# Scenario Cards, Selected (all 12 in the report)

场景卡精选 (12 卡全集见正文)

## S01 Low-speed delivery robots

S01 低速配送机器人

L4 + L2 time-share · 2 y · human fallback in snow

L4+L2 分时 · 2 年 · 雪雨天降级人工

## S08 Night cleaning (a pure time lease)

S08 夜间清扫 (纯时间租约)

Leases the spine's night hours — time is a layer

租的是主脊的夜间时段——时间也是层

## S09 Waste-heat interface (classification demo)

S09 算力余热接口 (判层示范)

Looks AI, files as L4 → a 10-year term

貌似 AI 实属 L4 → 10 年租期

## S12 Annual Chassis Day

S12 年度底盘日

One hour of L5 silence audits the parity paths

全带 L5 静默 1 小时，审计等价路径

All deposits illustrative; a non-AI parity path is statutory in every lease.



# Projects and Phasing: R-01 ∙∙ R-10

项目与分期：R-01 ∙∙ R-10

## 一期 0–1 年 Phase 1

零施工 zero construction

R-01 plans · R-02 census · R-05 lease system · R-07 curation · R-10 baseline

R-01 图则 · R-02 颗粒普查 · R-05 租约系统 · R-07 策划 · R-10 基线档案

## 二期 1–5 年 Phase 2

可逆 reversible

R-04 demo street · R-08 paving 1 km · R-09 conduit assessment

R-04 示范街 · R-08 年表带一期 · R-09 管廊评估

## 三期 5–25 年 Phase 3

择机 as opportunity allows

R-03 corridor redemption · R-06 grain pilot

R-03 线位/地役权兑现 · R-06 颗粒试点

## 四期 25 年+ Phase 4

制度 institutional

Assembly cycle: 2051 · 2076 · 2101 · 2126

复线大会周期运转：2051 · 2076 · 2101 · 2126

Every project carries an exit; chains  $\leq 2$ ; R-10 seals this open call as the 2051 control group.

# Metrics and Compliance

## 指标与合规

- First metric: slow\_layer\_change\_count = 0, always
- site\_area 11,412,825 m<sup>2</sup> · green 21.6% · public 4.0% (provisional)
- Computed: grain baseline (R-02, 2026-08-21 OSM); pending: conduit headroom (R-09), operations
- Coverage: announcement 1.3/1.4/1.5 + agent.1–agent.6
- Provisional boundary flagged in five places
- All spatial conclusions are conceptual recommendations

### 指标—证据链 Metrics—Evidence Chain

| 23 核心指标 · 已知/待复算 Core indicators · known vs pending                                                                                                     |              |   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---|
| slow_layer_change_count                                                                                                                                 | 0 count      | ● |
| reserved_row_count                                                                                                                                      | 4 count      | ● |
| reserved_row_length_m                                                                                                                                   | 13150.9 m    | ● |
| easement_passage_count                                                                                                                                  | 2 count      | ● |
| easement_coverage_ratio                                                                                                                                 | 待复算 pending  | ● |
| superblock_area_share                                                                                                                                   | 0.8478 ratio | ● |
| median_block_area_ha                                                                                                                                    | 0.662 ha     | ● |
| superblock_count                                                                                                                                        | 66 count     | ● |
| ew_crossing_count                                                                                                                                       | 8 count      | ● |
| spine_to_corridor_offset_m                                                                                                                              | 1247.2 m     | ● |
| device_band_coverage_km                                                                                                                                 | 1.565 km     | ● |
| corridor_conduit_headroom_years                                                                                                                         | 待复算 pending  | ● |
| section_redundancy_ratio                                                                                                                                | 待复算 pending  | ● |
| lease_active_count                                                                                                                                      | 待复算 pending  | ● |
| lease_return_rate                                                                                                                                       | 待复算 pending  | ● |
| lease_default_exit_rate                                                                                                                                 | 待复算 pending  | ● |
| deposit_pool_balance                                                                                                                                    | 待复算 pending  | ● |
| non_at_parity_coverage                                                                                                                                  | 1 ratio      | ● |
| annual_silent_drill_completion                                                                                                                          | 待复算 pending  | ● |
| centennial_review_interval_years                                                                                                                        | 25 years     | ● |
| timeline_paving_length_m                                                                                                                                | 1000 m       | ● |
| speed_diversity_index                                                                                                                                   | 待复算 pending  | ● |
| component_library_items                                                                                                                                 | 6 count      | ● |
| ● 实心 = 已知 (几何/设计声明/OSM 普查复算) ○ 灰 = 待正式数据/运营数据<br>filled = known (geometry / design declaration / OSM census) · grey = pending official/operational data |              |   |

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### 证据链 Evidence chain



### R-02 街坊颗粒普查协议 Block-grain census protocol



### 三项视觉核验指标 Three visual-gate metrics



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